

Arman Javan Sekhavat

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EDUCATION

Bachelor's Degree, Mechanical Engineering

2020–2024

K. N. Toosi University of Technology

TEHRAN, IRAN

- GPA: 3.79/4 (17.95/20) – 3.81/4 (17.99/20) Last two years
- Thesis Title: Design of a Controller for a Robotic Bicycle Using Intelligent Control Techniques
- Thesis Grade: A (19.00/20)
- Supervisor: Dr. S. Hossein Sadati

High School Diploma, STEM

2017–2020

National Organization for Development of Exceptional Talents (SAMPAD)

RASHT, GUILAN

RESEARCH INTERESTS

- Robot control through deep reinforcement learning
- Mechatronics

TEACHING EXPERIENCE

Teaching Assistant, Automatic Control

Sep. 2023 – Jan. 2024

Mechanical Engineering Department, K. N. Toosi University of Technology

Presented by Dr. Naser Naserifar

- Conducted exercise-solving classes
- Produced educational materials

Teaching Assistant, Computer Programming

Feb. 2022 – Jun. 2022

Mechanical Engineering Department, K. N. Toosi University of Technology

Presented by Dr. Farschad Torabi

- Conducted exercise-solving classes focused on programming in C++ using the Qt framework
- Produced educational materials

TECHNICAL AND RESEARCH EXPERIENCE

- **Control of a Two-wheeled Mobile Robot**
Design of a controller for a two-wheeled mobile robot using pole-placement and LQR
- **Mobile Robot Control System**
Design of a PID-based control system for a mobile robot. A GUI application serves as a control panel. The user is able to draw the desired path of the robot in the control panel.
- **Passive Steering Wheel**
Design of a CNN model capable of estimating the steering angle from an input image of a steering wheel

TECHNICAL SKILLS

- Programming languages and mathematical packages: C, C++, MATLAB, Python
- Software libraries and frameworks: OpenCV, MuJoCo, MLPACK, Qt
- Computer aided design: SOLIDWORKS
- Development boards: ESP32, Arduino
- Operating systems: Windows
- Other: Deep RL, Simulink, Socket Programming, L^AT_EX

COURSES STUDIED

Course Name	Grade (out of 20)
Neural Networks	19.00
Modern Control	18.80
Automatic Control	17.34
Introduction to Robotics and Lab.	19.15
Introduction to Mechatronics	18.00
Measurement and Control Systems	18.50
Mechanisms Design	18.00
Engineering Drawing II	19.60
Mechanical Vibrations	20.00

CURRENT RESEARCH

Integrating deep reinforcement learning with linear control techniques to control nonlinear systems under uncertainties and disturbances

HONORS AND AWARDS

- **Ranked within the top 10% of mechanical engineering students from the same year of entrance**
Issued by K. N. Toosi University of Technology • Sep. 2024
- **Ranked within the top 1.5% in the Iranian University Entrance Exam**
Issued by National Organization of Educational Testing • Sep. 2020

LANGUAGES

- **English**
Professional working proficiency
- **Persian**
Native language

REFERENCES

Farschad Torabi, Associate Professor
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